HLTH-02 Real-Time Emergency Resource Allocator — API Directory

> **Base URLs:**
>
> - **Backend REST API:** `http://localhost:4000`
> - **Real-time WebSockets:** `ws://localhost:4000` (Socket.io)
> - **Python ML Forecaster (XAI):** `http://localhost:8000`

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1. 🧑‍🦱 Citizen / Caller (Emergency SOS App)

Method	Endpoint	Description
`POST`	`/api/v1/incidents/sos`	**Trigger SOS:** Submits GPS coordinates (`lat` / `lng`) & emergency description. Automatically runs Claude Haiku clinical triage and queues for hospital allocation.
`GET`	`/api/v1/incidents/:id`	**Live Incident Tracking:** Returns live status (`RESERVED`, `EN_ROUTE`), assigned hospital name, ambulance vehicle number, and 5-minute reservation hold timer.

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2. 🎧 EMS Dispatcher (Central Command Console)

Method	Endpoint	Description
`GET`	`/api/v1/dispatcher/active-incidents`	**Active Queue:** Returns all waiting emergency cases sorted by clinical triage urgency (`RESUSCITATION` first).
`POST`	`/api/v1/dispatcher/solve-batch`	**AI Batch Optimizer:** Evaluates all waiting patients simultaneously across Pune, optimizing travel times, capacities, and $P(\text{available at ETA})$.

Method	Endpoint	Description
`GET`	`/api/v1/dispatcher/hospitals`	**Hospital Overview:** Live bed, ICU, and ventilator capacities across all 6 hospitals. Automatically flags stale hospitals (> 15m since last heartbeat).
`GET`	`/api/v1/dispatcher/ambulances`	**Fleet Telemetry:** Live GPS coordinates, availability, and trip statuses for all ambulances.

3. 🏥 Hospital ER Staff & Doctors (ER Reception Terminal)

Method	Endpoint	Description
`GET`	`/api/v1/hospitals/:id/reservations`	**Inbound Cases:** Shows incoming ambulances, patient clinical notes, reserved unit numbers, and hold countdown timers.
`POST`	`/api/v1/hospitals/reservations/:resId/accept`	**1-Click Accept:** Confirms bed hold, alerting driver and dispatcher that ER is ready.
`POST`	`/api/v1/hospitals/reservations/:resId/reject`	**Reject & Auto-Replan:** Rejects reservation; backend instantly re-optimizes the in-transit ambulance to the next best hospital in < 1 second.
`POST`	`/api/v1/hospitals/reservations/:resId/handover`	**Complete Handover:** Patient is admitted to bed; frees ambulance back to the available pool.
`PATCH`	`/api/v1/hospitals/:id/status`	**Temporary Shutdown / Re-open:** Closes hospital (e.g. power cut / ER diversion).

Method	Endpoint	Description
		Automatically reroutes all inbound ambulances!
`POST`	`/api/v1/hospitals/:id/resources`	**Add Discrete Units:** Adds new physical equipment (e.g. `+2 VENTILATOR`, `+1 ICU_BED`).
`PATCH`	`/api/v1/hospitals/:id/resources/adjust`	**Quick Capacity Delta:** Adjusts available units with delta (`+1` to free up, `-1` to mark under maintenance).
`POST`	`/api/v1/hospitals/:id/heartbeat`	**Heartbeat Refresh:** Updates last-reported timestamp and clears the 15-minute staleness warning penalty.

4. 🚑 Ambulance Paramedic / Driver (Navigation Tablet)

Method	Endpoint	Description
`PATCH`	`/api/v1/ambulances/:id/location`	**GPS Telemetry Ping:** Updates driver's live latitude and longitude coordinates on the dispatcher map.
`POST`	`/api/v1/ambulances/:id/status`	**Trip Status Transition:** Updates status (`DISPATCHED` $\rightarrow$ `EN_ROUTE` $\rightarrow$ `ARRIVED`).

5. 🧠 AI & Machine Learning Service (Decision Intelligence)

Method	Endpoint	Description
`POST`	`http://localhost:8000/predict`	**Explainable AI (XAI) Forecaster:** Calculates calibrated $P(\text{available at ETA})$ with feature

Method	Endpoint	Description
		attributions (bed buffer $+18\%$, peak hour -7%, travel delay -4%) and clinical summary.
`GET`	`http://localhost:8000/health`	**ML Service Health:** Verifies the Calibrated HistGradientBoosting model pipeline artifact is loaded in memory.

6. 🏆 Judges & Evaluators (Benchmark Scorecards & Stress-Tests)

Target	Command / Endpoint	Description
API	`GET /api/v1/evaluator/benchmark`	**Proof Scorecard:** Live comparative table between Naive Baseline and Proposed Smart Engine (Success rate, Reroutes, Double-booking collisions).
CLI	`npm run test:concurrency`	**50-Thread Race Attack:** Proves zero double-booking under extreme simultaneous caller load.
CLI	`npm run test:benchmark`	**Scenario Replay:** Deterministically verifies all 5 mandatory hackathon scenarios.
CLI	`python evaluate_model.py`	**ML Benchmark:** Proves +40.4% Brier Score improvement on held-out test data over baseline.